

Exploring the Impact of AI Skills on Job Security Perceptions Among SME Employees: A Mixed-Methods Analysis

Viktória Kubišová

This manuscript was compileD on November 1, 2024

Abstract

This study explores the relationship between AI skills and job security perceptions among SME employees. Using a mixed-methods approach, the research reveals that higher AI proficiency and confidence in AI usage correlate positively with job security and satisfaction. Employees frequently using AI and participating in company-sponsored training report enhanced job security, though those with advanced skills are more aware of AI's potential for job displacement. Demographic factors, such as age, education, and industry, also influence AI perceptions, with younger, tech-focused employees reporting stronger security. The findings underscore the importance of targeted AI training for SMEs to build employee confidence and adaptability. Limitations include sample demographic constraints; future research should broaden these insights through diverse, longitudinal studies across organizational contexts.

Keywords: Artificial Intelligence (AI), Job Security Perceptions, Small and Medium-sized Enterprises (SMEs), AI Skills, Workforce Adaptability, AI Training, Employee Confidence, Mixed-Methods Research, Technological Change, Organizational Support

E-mail address: 231781@buas.nl

1. Introduction

The rapid integration of artificial intelligence (AI) technologies in business operations is transforming workplaces, particularly for small and medium-sized enterprises (SMEs). While AI has the potential to automate routine tasks and increase efficiency, its adoption has raised concerns regarding job security, especially among employees who may lack the skills to adapt to AI-driven changes [7]. In SMEs, where resources for upskilling are often limited, employees' perceptions of job security may hinge on their ability to leverage AI effectively. This study aims to investigate the relationship between SME employees' AI skills and their belief that these skills contribute positively to their job security, thereby insights into AI's impact on workforce dynamics within smaller organizations.

This research will address not only the main question of how AI skills relate to employees' perceptions of job security but will also explore related factors, such as the frequency of AI tool usage, employee confidence in using these tools, and the impact of company-provided AI training. Additionally, demographic influences on AI perceptions and the perceived risk of job displacement will be examined to provide a nuanced view of how AI impacts job security across different employee groups.

1.1. Research Questions and Hypothesis

1.1.1. The Main Focus of the Research

The main focus of this research is to explore how SME employees' skills in using AI tools relate to their perceptions of job security. Given the growing integration of AI in workplaces, understanding the link between employees' AI skills and their sense of job stability is critical for developing a deeper insight into how technology impacts workforce dynamics in SMEs.

Main Research Question:

- How does having the skills to use AI tools relate to SME employees' perceptions of job security?

Hypotheses:

- Null Hypothesis (H₀):** Having the skills to use AI tools does not significantly influence SME employees' perceptions of job security.
 - This hypothesis suggests that possessing AI-related skills has no meaningful impact on employees' sense of job se-

curity in SMEs, regardless of technological advancements brought by AI in the workplace.

- Alternative Hypothesis (H₁):** Having the skills to use AI tools significantly influences SME employees' perceptions of job security.
 - This hypothesis suggests that employees with AI-related skills perceive higher job security compared to those without such skills, as these skills are seen as valuable in adapting to AI integration within the workplace.

1.1.2. Subquestions

To fully understand and explore the main research question, it is essential to investigate related factors that could influence the outcomes. Therefore, we will also address several sub-questions that provide further context and enrich our analysis of how AI skills impact job security in SMEs:

- How do the frequency of AI tool usage and employees' confidence in using these tools contribute to their perceptions of job security?**

This sub-question will combine the exploration of how often employees interact with AI tools and their confidence in using them, aiming to determine whether frequent and confident use of AI correlates with higher job security.
- How do AI skills and perceptions of job security vary across different demographics (e.g., gender, age, role, industry)?**

This sub-question seeks to explore whether demographic factors, such as age, gender, industry sector, or job role, influence the relationship between AI skills and job security. It will provide insights into whether specific groups feel more or less secure due to AI adoption.
- What role does AI training provided by the company play in shaping employees' perceived job security?**

This question focuses on how organizational support through formal AI training programs affects employees' perceptions of job security. The study will explore whether employees who receive AI training feel more secure compared to those who do not.
- How does the perceived risk of job displacement due to AI adoption relate to employees' AI skills and their confidence in using AI tools?**

This sub-question will explore the concept of job risk rather than

security, investigating whether employees with AI skills and confidence in using AI tools feel less at risk of job displacement compared to those without such skills.

2. Literature Overview

The adoption of AI in workplaces has led to varied employee responses, often dependent on AI proficiency. Employees skilled in AI view it as a chance for growth, while those without AI skills tend to see it as a threat to job stability, feeling more vulnerable to automation-related job displacement. [1], [2] These differing perspectives highlight AI proficiency as a key factor in workforce attitudes toward technology.

In SMEs, AI adoption challenges are amplified by limited resources, often hindering comprehensive training programs. Research indicates that SMEs may struggle to offer sufficient upskilling, leaving employees unprepared for technological changes in their roles. [6] However, companies that invest in AI skill-building see more positive workforce attitudes, as trained employees feel secure and adaptable. [3] This link between AI skills and job security is particularly strong in SMEs, where employees with AI competencies are more likely to view technological change as beneficial. [7]

Demographic factors also influence AI perceptions, with younger employees and those in tech-focused sectors seeing AI as an opportunity, while older employees perceive it as a threat. [5] Confidence in AI usage enhances job security, as frequent users feel less vulnerable to automation. [4] Organizational support through AI training also boosts adaptability and job security, underscoring the importance of company-provided resources in preparing employees for AI's impact. [3]

Despite existing research, there remains a gap in understanding how AI-related skills specifically impact job security in SMEs, where training resources are limited. [5] While larger corporations offer robust AI training, SMEs often face distinct challenges in supporting employees' adaptation to AI. [6] Existing studies tend to focus on large organizations, offering limited insight into SME employees' perceptions of job stability in relation to AI skills and confidence. [3]

Addressing this gap, this study explores how AI skills, training access, and demographic factors shape job security perceptions among SME employees, aiming to provide insights tailored to smaller businesses integrating AI with fewer resources.

3. Methodology

This study utilizes a **mixed-methods research design**, combining quantitative and qualitative data to investigate how AI skills relate to SME employees' perceptions of job security. This dual approach provides both broad statistical patterns and deeper individual insights.

3.1. Study Population and Sample

The study focuses on SME employees, specifically those who have experience or interactions with AI tools in their work environment. For quantitative data, participants were recruited through **simple random sampling** on Prolific, focusing on individuals aged 18 or older. While the survey primarily aimed to capture responses from SME employees, any responses from larger organizations (over 250 employees) were filtered out to maintain the SME focus.

For qualitative data, **convenience and purposive sampling** were employed. Contacts for follow-up interviews with SME owners were initially gathered at an AI-event hosted by Digiwerkplaats. Additional recruitment efforts included reaching out to local SMEs in Breda; however, due to limited responses, further contacts were established through employees at the SME company **VEED**. This dual approach enabled the collection of broad quantitative data and focused qualitative insights from SME employees/owners.

3.2. Data Collection Methods

3.2.1. Quantitative Data Collection

Quantitative data were collected through an online Qualtrics survey administered via Prolific, developed collaboratively by three research teams focused on AI implementation in SMEs. The survey consisted of 34 Likert-scale, multiple-choice, and open-ended questions covering AI skill level, job security perceptions, confidence in AI tool use, and demographics (age, profession, gender, and education). Participants reported their AI tool usage frequency, types of tools used, any employer-provided AI training, and other relevant variables. The survey took approximately 10–15 minutes to complete, and responses were securely stored in .csv format on a university server.

3.2.2. Qualitative Data Collection

Qualitative data were gathered via semi-structured interviews with selected survey participants to explore their experiences and attitudes toward AI in the workplace. Open-ended questions addressed job performance, satisfaction, workplace optimization, views on AI's role in job security, and ethical concerns. Conducted over Microsoft Teams, each 30–40-minute interview was recorded and transcribed for analysis.

3.3. Data Analysis Techniques

3.3.1. Quantitative Data Analysis

Quantitative data from the survey were analyzed using a combination of visual exploration, statistical inference, and hypothesis testing. Given that the primary variables were categorical, traditional descriptive statistics like means and standard deviations were not commonly used. Instead, data visualization techniques were employed, including grouped and stacked bar charts, heatmaps, violin plots, box plots, swarm plots, and facet grid plots, allowing for effective examination of single-variable (univariate), two-variable (bivariate), and three (or more)-variable (multivariate) relationships.

Encoding Key Variables

Prior to statistical analysis, key variables were numerically encoded to establish an ordinal order based on survey questions. For AI knowledge levels, participants responded to the question, **"How would you describe your current knowledge of AI?"** Responses ranged from **Level 1** ("no knowledge of AI") to **Level 5** ("can develop AI models using frameworks like Tensorflow or PyTorch"). Job security perceptions were similarly encoded, in response to the question, **"Do you believe your ability to use AI tools contributes positively to your job security?"** with levels ranging from **Level 1** ("Definitely not secure") to **Level 5** ("Definitely secure").

Hypothesis Testing and Correlational Analysis

Hypothesis testing was conducted using **ANOVA** and a **T-test** to explore the link between AI knowledge levels and job security perceptions:

• ANOVA Approaches:

- **Excluding Small Groups:** The lowest and highest knowledge levels (Levels 1 and 5) were excluded to prevent skewed results due to insufficient group sizes, focusing on Levels 2, 3, and 4 for clearer statistical insights.
- **Revised Grouping:** Levels 1 and 2 were combined as "Lower AI Knowledge," Level 3 was retained as "Moderate AI Knowledge," and Levels 4 and 5 were merged into "Higher AI Knowledge."

- **T-Test:** A binary grouping compared job security perceptions between Higher AI Knowledge (Levels 3–5) and Lower AI Knowledge (Levels 1–2), assessing statistical significance.

Correlation analysis was also conducted to examine relationships between AI skill level, frequency of AI usage, confidence in AI usage, job risk, job security perceptions, job satisfaction, comfort in AI usage,

and AI training offered by the company. All analyses were conducted in Python, with detailed documentation in the Jupyter Notebook.

3.3.2. Qualitative Data Analysis

Qualitative data from the interviews were analyzed using thematic analysis to identify patterns and recurring themes related to job security perceptions and AI skill levels. The process included transcription, familiarization, coding, and theme development. Transcriptions were read and re-read, with meaningful text segments highlighted and labeled using a color-coding system shown below, where related codes shared similar shades. Similar codes were grouped into broader themes, with sub-themes capturing nuances. Each interview was summarized to outline its key topics and themes. This structured approach provided deeper insights into employees’ attitudes toward AI and their perceived job stability, adding depth to the quantitative findings.

Themes and subthemes	
1. Operational Efficiency	
AI Tool Integration:	Use of AI tools like ChatGPT, Miro, and Cursor for streamlining tasks.
Task Automation and Simplification:	AI’s role in automating repetitive tasks and improving workflow efficiency.
Challenges with AI Use:	Limitations, technical issues, or initial learning curves encountered with AI tools.
Skill Awareness and Guidance:	Training needs for effective AI use, especially in prompt crafting.
2. Job Satisfaction	
AI as a Task Enhancer:	AI’s supportive role in making specific tasks easier without fundamentally altering job satisfaction.
Mixed Reactions to AI Push from Management:	Positive and negative feelings about management expectations to use AI, impacting job satisfaction.
3. Job Security and Stability	
AI as a Support, Not a Threat:	AI skills enhance roles but are not seen as replacements for human expertise.
Growing Standard for AI Skills:	AI skills are becoming expected but not the only factor in job security.
4. Ethics and Privacy in AI Use	
Data Sensitivity and Privacy Concerns:	Privacy issues, especially with user data in AI tools.
Bias and Ethical Implications:	Awareness of ethical issues and biases in AI, especially in creative applications.
5. Creativity and AI’s Role in Innovation	
AI as a Creative Aid:	AI as a tool to support creativity, providing ideas or options without replacing human input.
Balance Between AI and Human Input:	Importance of maintaining originality and human oversight in creative work.

Figure 1. Color Codes Used for the Qualitative Analysis

3.4. Ethical Considerations

The study adhered to ethical research standards through several protocols. Participants received a research information letter explaining the study’s purpose, data usage, and their rights, and they provided informed consent before participating. To ensure confidentiality and anonymity, data were anonymized and assigned unique identifiers, with only anonymized data analyzed. All files were securely stored on encrypted university servers, accessible solely to the research team. Data storage complied with GDPR and institutional policies, with data securely stored on a GitHub repository for the project’s duration and retained for ten years for potential future research use.

4. Results

4.1. Overview of the Survey and Interview Sample

Sample Demographics

The sample included 154 SME employees with diverse backgrounds in age, gender, education, nationality, and company background. Most respondents were aged 18–44, with a higher representation of females (101) compared to males (46). Education levels varied, with the majority holding a Bachelor’s degree (84), followed by those with a Master’s degree (31). Participants primarily worked in SMEs with up to 100 employees and revenue below €5 million. Geographically, the sample was dominated by respondents from South Africa (84), followed by the United Kingdom (12).

Key Variables

- AI Knowledge:** The majority (113 participants) had basic AI awareness, primarily using tools like ChatGPT without coding skills. A smaller group (19) could use AI APIs via code, while only 1 respondent had advanced AI model development skills, indicating basic to intermediate AI literacy across the sample.

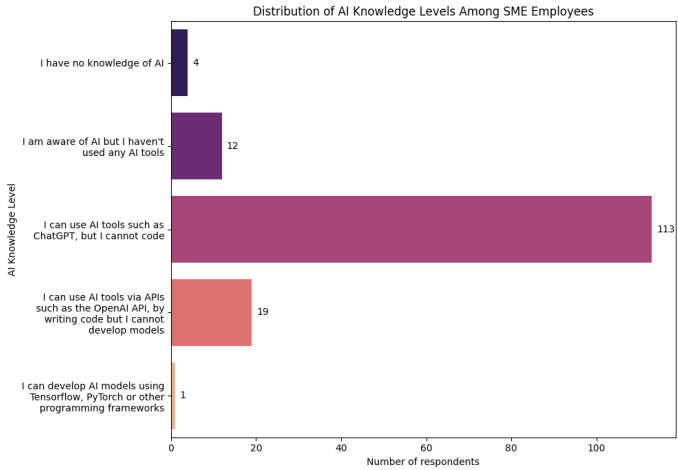


Figure 2. Distribution of AI Knowledge Levels Among SME Employees

- Job Security Perceptions:** Perceptions of AI’s impact on job security were varied. While 51 respondents felt that their ability to use AI tools “probably” contributed positively to their job security, 54 were uncertain about AI’s role in enhancing job stability.

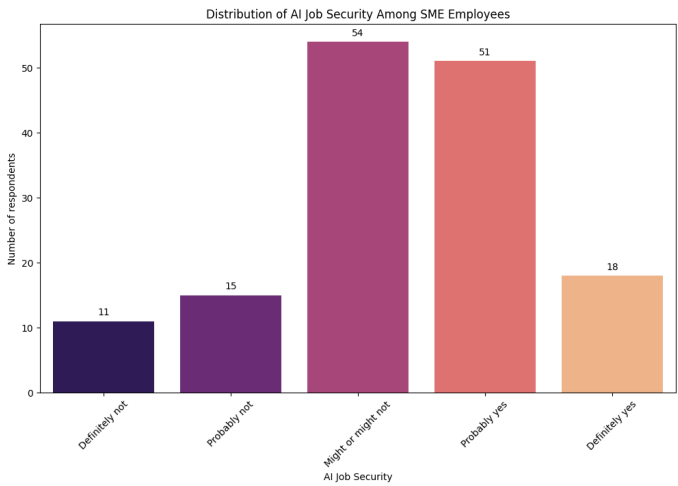


Figure 3. Distribution of AI Job Security Among SME Employees

4.2. Main Research Question: Relationship Between AI Skills and Job Security Perceptions

Summary of Analysis

The analysis examines the relationship between AI skill levels (AI_Knowledge_Level1) and SME employees’ perceptions of job security (AI_Job_Security). With increasing AI adoption, this study assesses whether employees’ ability to use AI tools contributes positively to their sense of job security. Descriptive and inferential statistics were used to compare job security perceptions across AI knowledge levels, identifying significant differences among groups and exploring how proficiency relates to confidence in job stability.

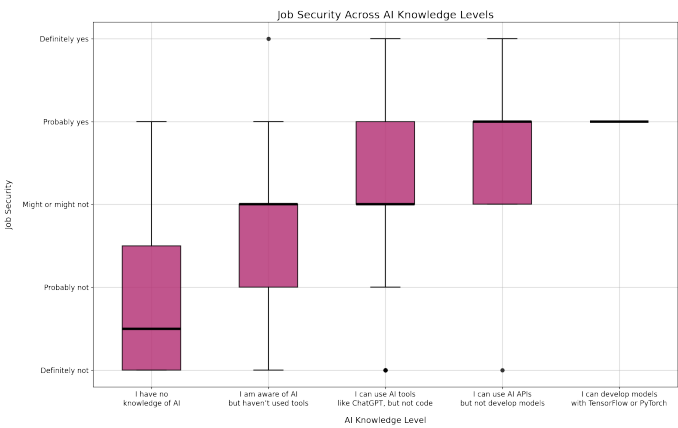


Figure 4. Box Plot of the Distribution of AI Job Security Perceptions Across AI Knowledge Levels Among SME Employees

Descriptive Findings

The analysis of job security perceptions across AI knowledge levels shows a clear trend: higher AI proficiency aligns with increased job security among SME employees. The box plot above illustrates the distribution of job security perceptions, with higher AI knowledge levels exhibiting a narrower spread and generally higher median job security scores, suggesting greater confidence in job stability as AI proficiency rises. This trend is further supported by the bar plot below, which illustrates a progressive increase in mean job security perception from lower to higher AI knowledge levels.

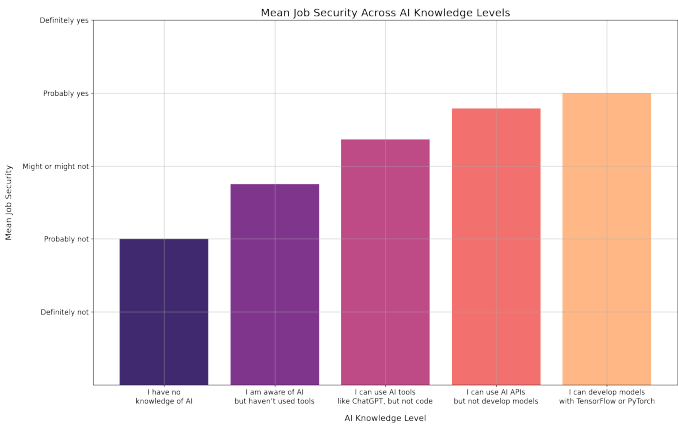


Figure 5. Bar Chart of the Mean AI Job Security Perception for AI Knowledge Levels Among SME Employees

Additional analyses, including grouped bar charts, such as Figure 6, and heatmaps, indicate that employees with skills such as using AI tools like ChatGPT (without coding) or utilizing APIs like the OpenAI API tend to feel more secure in their jobs, with many respondents in these categories noting that their AI skills “probably” or “definitely” enhance job security. Findings from a violin plot further reinforced this pattern, showing that as AI knowledge increases, job security perceptions become not only more positive but also more consistent.

4.3. Statistical Results

To test the hypothesis regarding the relationship between AI knowledge and job security perceptions, three analytical approaches were used: two ANOVA tests with different groupings and an independent samples T-test.

Approach 1: One-Way ANOVA Excluding Small Groups

The first ANOVA excluded the smallest groups (Levels 1 and 5) to focus on Levels 2, 3, and 4, which had adequate sample sizes. Levene’s test confirmed homogeneity of variances ($p = 0.69$), allowing the

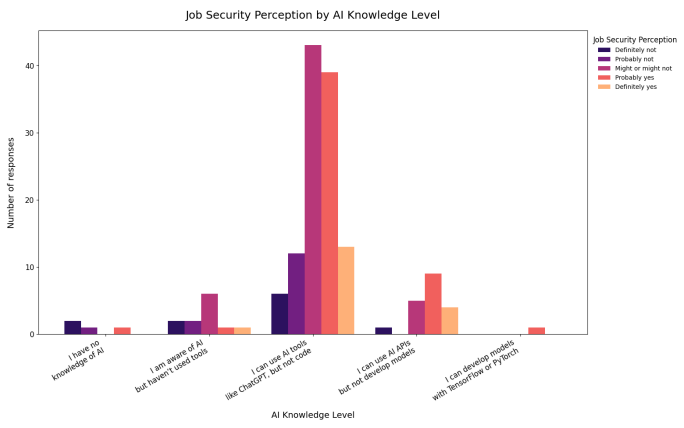


Figure 6. Grouped Bar Chart of Job Security Perceptions by AI Knowledge Levels Among SME Employees

analysis to proceed despite minor normality deviations in Levels 3 and 4. The ANOVA revealed a significant difference in job security perceptions across the knowledge levels of our focus ($F = 3.91, p = 0.0222$). Tukey’s HSD post-hoc test showed that Level 4 respondents, who could use APIs but not develop models, reported higher job security perceptions than those in Level 2 ($p = 0.0162$), but no other significant pairwise differences were found.

Approach 2: One-Way ANOVA with Revised Grouping

The second ANOVA consolidated Levels 1 and 2 into a “Lower AI Knowledge” group, retained Level 3 as “Moderate AI Knowledge”, and combined Levels 4 and 5 as “Higher AI Knowledge” to create larger, more stable categories. Homogeneity of variances was again confirmed ($p = 0.379$), though normality was not met in all groups. Results indicated a significant difference among the revised groups ($F = 6.74, p = 0.0016$). Tukey’s HSD test revealed that both Moderate and Higher AI Knowledge groups had significantly higher job security perceptions than the Lower group, with no significant difference between Moderate and Higher groups.

Approach 3: Independent Samples T-Test

For a broader comparison, Levels 1 and 2 were grouped as Lower AI Knowledge, while Levels 3–5 were grouped as Higher AI Knowledge. Welch’s t-test was conducted to account for unequal variances and sample sizes, confirming that employees with Higher AI Knowledge felt significantly more secure than those with Lower AI Knowledge ($t = -2.75, p = 0.0133$), with a Cohen’s d of 0.78, indicating a moderate to large effect size.

These findings consistently demonstrate a positive relationship between AI skills and job security perceptions, with moderate proficiency in AI yielding a substantial increase in perceived job security. Further skill gains, however, did not significantly enhance perceptions beyond this moderate level.

4.4. Impact of AI Tool Usage Frequency and Confidence on Job Security and Job Satisfaction

Employees with intermediate to advanced AI skills, particularly those confident in their abilities, tend to use AI tools frequently. These users often associate their AI skills with job security, frequently choosing positive security perceptions and reporting high job satisfaction. In contrast, employees with minimal skills or low confidence rarely engage with AI, feel less secure, and show varied satisfaction, underscoring the importance of proficiency and confidence for stability and satisfaction. A strong link between AI confidence and job security/satisfaction is evident, with higher confidence correlating with more positive outcomes.

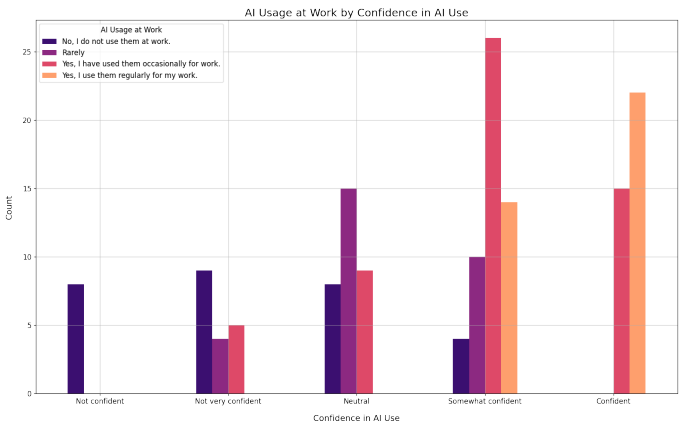


Figure 7. Grouped Bar Chart of AI Usage at Work by Confidence in AI Usage Among SME Employees

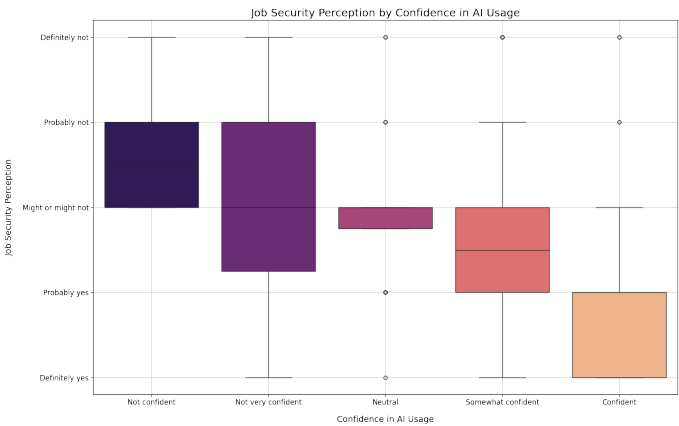


Figure 8. Box plot of Job Security Perceptions by Confidence in AI Usage Among SME Employees

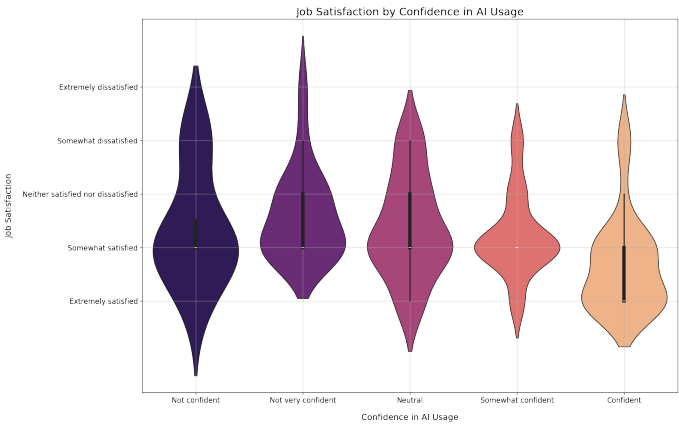


Figure 9. Violin plot of Job Satisfaction Levels by Confidence in AI Usage Among SME Employees

4.5. Variations in AI Skills and Job Security Perceptions Across Demographics

Basic AI familiarity, such as ChatGPT usage, is prevalent among younger employees, those in mid-sized firms, and individuals with Bachelor’s degrees, while advanced skills are less common. Employees in tech-focused roles and industries (e.g., Software, Consulting) exhibit higher AI proficiency than those in traditional sectors like Retail. Gender differences reveal higher AI skills among males. Job security perceptions are stronger among younger, educated employees in adaptable roles, especially in tech sectors, while older employees, females, and those in traditional roles express more uncertainty.

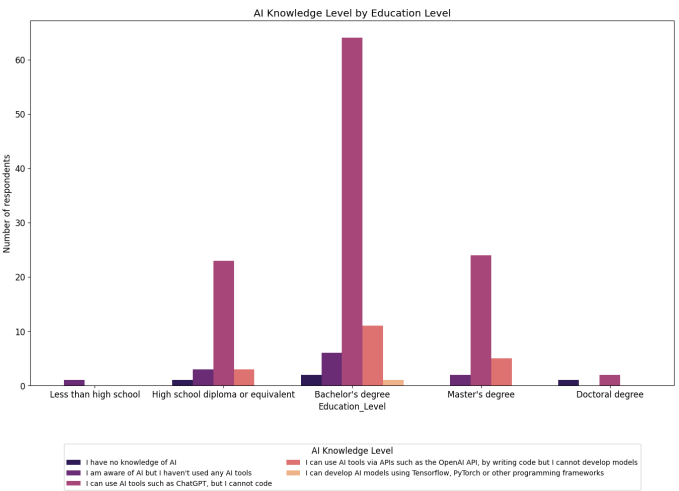


Figure 10. Bar plot of AI Knowledge Levels by Education Levels Among SME Employees

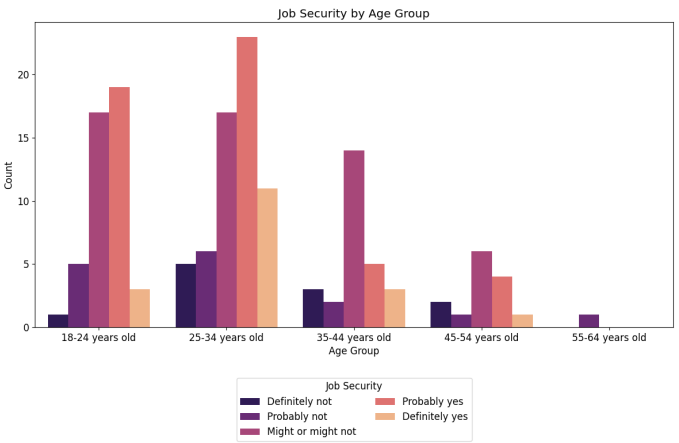


Figure 11. Grouped Bar Chart of Job Security Perceptions by Age Groups Among SME Employees

4.6. Impact of AI Training on Job Security Perceptions

Employees who participated in company-funded AI training report higher job security, often selecting positive security responses, while those without training express uncertainty. Employees offered but not participating in training show moderate security, suggesting that opportunity alone is less impactful. Training participants report greater confidence, comfort with AI, and higher satisfaction, whereas non-participants display lower confidence and more varied satisfaction levels.

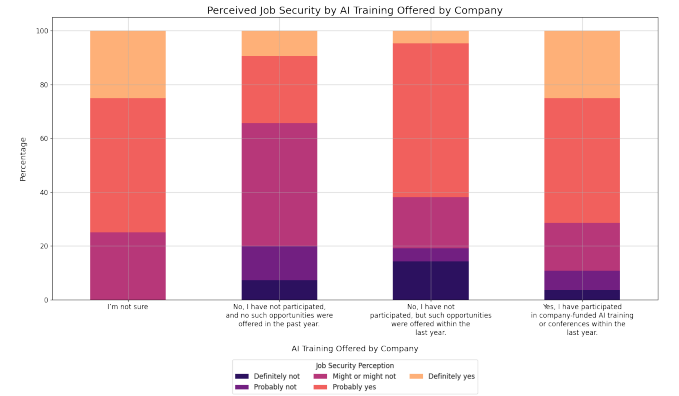


Figure 12. Proportional Stacked Bar Chart of Job Security Perceptions by AI Training Offered by Company Among SME Employees

4.7. Perceived Job Risk Due to AI and Its Relation to AI Skills

Employees with advanced AI skills perceive a higher risk of job replacement by AI, likely due to a clearer understanding of AI’s potential. However, higher confidence mitigates job displacement fears, as confident users report lower perceived risk. Perceived risk does not consistently affect job security or satisfaction, with varied responses across risk levels. Generally, employees who view AI as less threatening report higher satisfaction, though both high- and low-risk perceivers show a range of satisfaction levels.

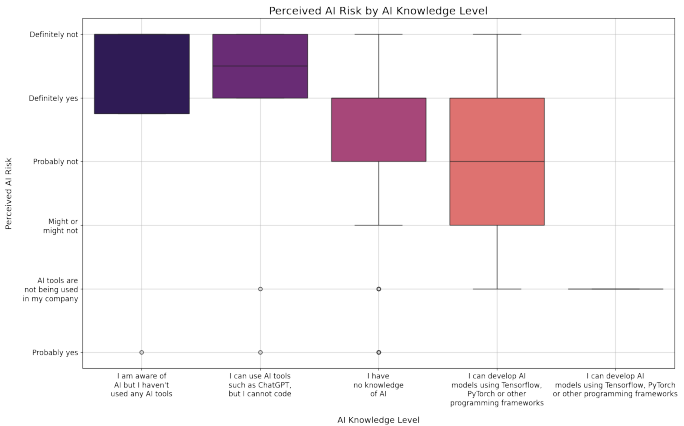


Figure 13. Bar Chart of Perceived AI Risk by AI Knowledge Levels Among SME Employees

4.8. Qualitative data analysis

The qualitative data analysis reveals several core insights into SME employees’ experiences and attitudes toward AI in the workplace. Across various roles, AI is perceived primarily as a productivity enhancer, aiding in tasks like document organization, idea clustering, and routine problem-solving, particularly in repetitive or administrative work. Notably, employees view AI as a tool that complements rather than replaces their skills, particularly in technical and creative fields where critical thinking and complex decision-making are essential. While AI contributes positively to job satisfaction by easing mundane tasks, some employees express discomfort with the pressure to adopt AI, feeling it sets higher expectations without necessarily enhancing satisfaction. Ethical considerations also emerge prominently, with concerns around data privacy, user consent, and the potential biases in AI deployment, especially in creative and sensitive data tasks. Although opinions on AI’s impact on job security vary, there is a general sense that roles requiring expertise and strategic insight are secure, with AI more likely to augment rather than replace these roles. Overall, the data reflect a cautious optimism toward AI’s potential, balanced by a critical awareness of its limitations and ethical implications.

5. Discussion

5.1. Relationship Between AI Skills and Job Security Perceptions

This study found a positive link between AI skills and job security among SME employees, with higher proficiency correlating with greater job security perceptions. This is supported by qualitative findings, where employees view AI as enhancing productivity and complementing, rather than replacing, human skills. [3], [6] However, advanced AI users are also more aware of AI’s potential for job displacement, indicating both an opportunity for enhanced job security and a heightened awareness of risk. These findings reinforce the importance of AI-related training in SMEs to support employee adaptability in the face of technological change.

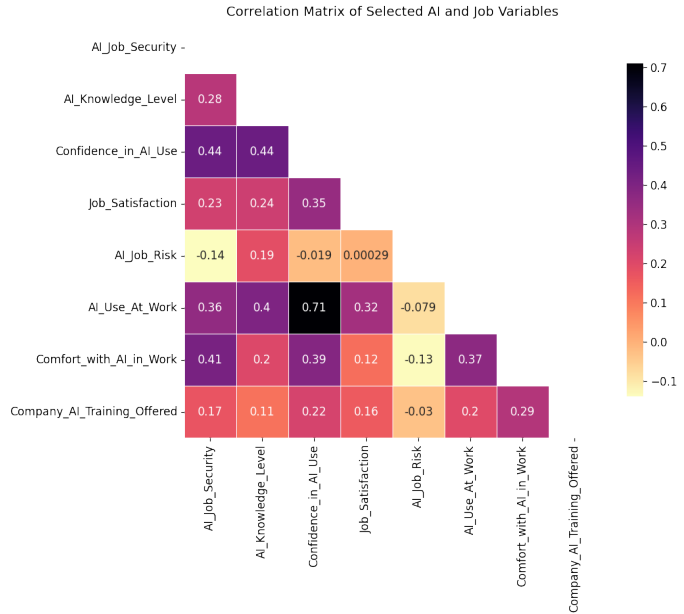


Figure 14. Correlation Matrix of Selected AI and Job Variables Examined Among SME Employees

5.2. Influence of Confidence and Frequency of AI Tool Usage

Frequent and confident AI users reported higher job security and job satisfaction, viewing AI as a beneficial asset. Qualitative data confirm that using AI for routine tasks boosts satisfaction by allowing employees to focus on strategic work. However, some employees feel pressured by management to adopt AI, highlighting a need for supportive environments that encourage voluntary engagement with AI tools. This emphasizes the role of training and mentorship in making AI use empowering rather than burdensome. [2], [4]

5.3. Demographic Influences on AI Skills and Job Security Perceptions

Younger, educated employees in tech roles reported stronger job security perceptions related to AI, while senior and non-technical employees showed more reservations. This pattern, evident in both data sets, suggests that targeted training for older and less technically skilled employees could bridge security perception gaps and reduce anxieties around AI adoption. Tailored support for these groups would foster a more resilient and inclusive workforce. [5]

5.4. Impact of Company-Provided AI Training

AI training significantly enhanced job security perceptions and satisfaction, with active participation proving more impactful than mere availability. Qualitative insights highlight that employees who received training felt more confident and less anxious about potential job displacement, with junior employees particularly valuing this upskilling opportunity for career advancement. These findings underscore the importance of engaging, accessible training formats to encourage active participation in AI education. [3]

5.5. Perceived Job Risk Due to AI and Its Relation to AI Skills

Employees with advanced AI skills reported a higher perceived risk of job displacement, but this was mitigated by confidence in AI use, which enhanced their sense of security. The qualitative data support this, indicating that knowledgeable employees view AI as a tool that complements their roles, especially in tasks requiring creativity. Organizational support that builds both AI skills and confidence can help employees balance the risks and benefits of AI adoption. [1], [3]

5.6. Reflection on the Integration of Qualitative and Quantitative Findings

Overall, the qualitative and quantitative findings are complementary, reinforcing the positive impact of AI skills on job security and satisfaction. The qualitative data add nuance by revealing feelings of pressure and ethical concerns around AI adoption, which were less evident in the quantitative results. These insights suggest that while AI can enhance job security, SMEs must also address psychological and ethical considerations to ensure employees feel genuinely supported.

5.7. Limitations and Future Research Directions

The study’s sample, administered via Prolific and primarily representing South African SME employees, included over two-thirds female participants, which may limit generalizability due to cultural and gender influences on AI perceptions. Qualitative data had limited diversity, with most responses from employees within a single SME, introducing potential organizational bias. Statistical analysis faced challenges from uneven AI knowledge level distribution; while ANOVA’s variance assumption was met, the normality requirement was not, necessitating cautious interpretation.

Future research would benefit from a more balanced sample across AI knowledge levels, broader geographic representation, larger sample size, and exploration in larger organizations where resources and AI integration differ substantially. Longitudinal studies could also provide insights into the effects of AI skill development on job security perceptions over time.

5.8. Practical Implications for SMEs

The study’s findings suggest practical steps for SMEs aiming to harness AI without undermining employee job security. [3], [6] First, AI training programs should be made accessible and actively promoted to enhance both skills and confidence in AI use. Qualitative feedback suggests that training should be flexible and supportive, reducing pressure and enabling employees to integrate AI at their own pace. Second, SMEs should consider tailored support for employees in non-technical roles or traditional industries, who may feel more threatened by AI’s potential impact on job stability. Ensuring that AI tools are seen as productivity aids, rather than replacements, can alleviate these concerns. Lastly, fostering an organizational culture of adaptability and continuous learning can empower employees to view AI as a tool for growth, especially when their feedback and ethical concerns about AI’s use are acknowledged and addressed.

6. Conclusion

This study offers valuable insights into the relationship between AI skills and job security perceptions among employees in small and medium-sized enterprises (SMEs). Through a mixed-methods approach, we found that higher proficiency and confidence in AI use are positively correlated with perceptions of job security and satisfaction. Employees who frequently use AI tools and participate in company-provided training generally report greater job security, though those with advanced AI skills are also more aware of the potential risks of job displacement. The findings underscore the importance of accessible, targeted AI training for SMEs, as it fosters a sense of adaptability and security among employees, helping them navigate technological changes effectively.

The qualitative insights provide a nuanced perspective on how AI is perceived across roles and demographics. While AI is widely regarded as a productivity enhancer that complements existing skills, there is a prevailing sentiment that AI is unlikely to replace roles requiring critical thinking and creativity. Ethical concerns around AI usage, particularly regarding data privacy and the pressure to adopt AI, highlight the need for SMEs to foster a supportive and transparent AI adoption environment. Moreover, demographic factors—such as

age, industry, and job role—play a role in shaping employees’ AI perceptions, with younger, tech-oriented employees feeling more secure, and senior or non-technical employees expressing more reservations.

This research contributes to the understanding of AI’s impact on workforce dynamics in SMEs and highlights practical implications for management. By investing in flexible, inclusive AI training and addressing employees’ ethical concerns, SMEs can create a resilient, future-ready workforce. Future research should aim to expand these insights across diverse industries and geographical contexts, with longitudinal studies to assess the long-term effects of AI skill development on job security perceptions. As SMEs continue to integrate AI, fostering an organizational culture that values both technological advancement and employee security will be essential for sustainable growth and workforce satisfaction.

■ Disclosure

ChatGPT (OpenAI, 2024) was used solely to refine language and improve clarity in this study. All interpretations, findings, and conclusions are the authors’ own.

■ References

[1] M. Arntz, T. Gregory, and U. Zierahn, “The risk of automation for jobs in oecd countries: A comparative analysis”, *OECD Social, Employment and Migration Working Papers*, vol. 189, 2016. DOI: 10.1787/5jlz9h56dvq7-en. [Online]. Available: <https://doi.org/10.1787/5jlz9h56dvq7-en>.

[2] C. B. Frey and M. A. Osborne, “The future of employment: How susceptible are jobs to computerisation?”, *Technological Forecasting and Social Change*, vol. 114, pp. 254–280, 2016. DOI: 10.1016/j.techfore.2016.01.005. [Online]. Available: <https://doi.org/10.1016/j.techfore.2016.01.005>.

[3] J. E. Bessen, “Ai and jobs: The role of demand”, *NBER Working Paper*, no. 24235, 2018. DOI: 10.3386/w24235. [Online]. Available: <https://doi.org/10.3386/w24235>.

[4] M.-H. Huang and R. T. Rust, “Artificial intelligence in service”, *Journal of Service Research*, vol. 21, no. 2, pp. 155–172, 2018. DOI: 10.1177/1094670517752459. [Online]. Available: <https://doi.org/10.1177/1094670517752459>.

[5] D. M. West, *The Future of Work: Robots, AI, and Automation*. Washington, D.C.: Brookings Institution Press, 2018. [Online]. Available: <https://www.brookings.edu/book/the-future-of-work/>.

[6] M. Cubric, “Drivers, barriers, and social considerations for ai adoption in business and management: A tertiary study”, *Technology in Society*, vol. 62, p. 101 257, 2020. DOI: 10.1016/j.techsoc.2020.101257. [Online]. Available: <https://doi.org/10.1016/j.techsoc.2020.101257>.

[7] L. Maghfirah and H. Eni, “Understanding the role of ai skills on workforce resilience in smes”, *Journal of Applied Business Research*, vol. 40, no. 3, pp. 301–310, 2024. [Online]. Available: <https://www.clutejournals.com/Journals/index.php/JABR>.